

1. Let $(V, + \cdot)$ be a vector space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Define what is meant by the statement that the mapping $N : V \rightarrow \mathbb{R}$ is a norm on V .

It is important for you to know the definitions!

2. Let $(V, \|\cdot\|)$ be a normed space over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Set

$$d(u, v) := \sqrt{\|u - v\|} \quad \forall u, v \in V.$$

- (a) Show that d is a distance on V . (Hint: $\sqrt{s+t} \leq \sqrt{s} + \sqrt{t}$ for every $s, t \geq 0$.)
(Why is the hint true?)

- (b) Is the distance d induced by a norm on V ? Explain clearly your answer.
 Show that N is a norm on $\mathbb{R}^2$.

You need think about thhis carefully! If d is induced by some norm N on $\mathbb{R}$, then by definition

$$N(u - v) = d(u, v) \quad \forall u, v \in V \quad \Rightarrow \quad \underbrace{N(u) = d(u, 0_V) = \sqrt{\|u\|}}_{(*)} \quad \forall u \in V.$$

Check from $(*)$ that such an N does not satisfy the scalar multiplication property of a norm

$$N(\lambda u) = |\lambda|N(u) \quad u \in V, \quad \forall \lambda \in \mathbb{K} \text{ ???}$$

Give an example that does not satisfy this identity!

3. Let $N : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $N(x, y) := \int_0^1 |x + ty| \quad \forall (x, y) \in \mathbb{R}^2$.

Show that N is a norm on $\mathbb{R}^2$.

You should be able to do this!

4. Let $V = \mathcal{C}([0, 1], \mathbb{R})$, the space of continuous functions from $[0, 1]$ to $\mathbb{R}$. Set

$$\|f\|_1 := \int_0^1 |f(t)| dt \quad \text{and} \quad \|f\|_\infty = \sup_{0 \leq t \leq 1} |f(t)| \quad \forall f \in V.$$

- (a) Show that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on V . **(This is trivial!)**

- (b) Is the mapping $N(f) := \left| \int_0^1 f(t) dt \right|$ also a norm on V ? Explain your answer.

(Is the vanishing property of a norm satisfied here?)

(c) Let (f_n) be the sequence in V defined by $f_n(t) := t^n$ for any $t \in [0, 1]$.

(i) Find the pointwise limit of (f_n) .

if $t = 1$ then $f_n(t) = 1$ and if $t \in [0, 1)$, then $f_n(t) \rightarrow 0$. So the sequence (f_n) converges pointwise to the function

$$f(t) := \begin{cases} 0 & \text{if } t \in [0, 1), \\ 1 & \text{if } t = 1. \end{cases}$$

(ii) Does the sequence (f_n) converge in $(V, \|\cdot\|_\infty)$? Explain clearly your answer.

Here, you have been asked to say whether the convergence is uniform. Since, (f_n) is a sequence of continuous functions which converges pointwise to a discontinuous function, then the convergence is not uniform!!!

(iii) Does the sequence (f_n) converge in $(V, \|\cdot\|_1)$? Explain clearly your answer. (Hint: Find $\|f_n\|_1$).

Following the hint, we find that

$$\|f_n\|_1 = \int_0^1 |f_n(t)| dt = \int_0^1 t^n dt = \frac{1}{n+1} \Rightarrow \lim_{n \rightarrow \infty} \|f_n\|_1 = 0.$$

Hence, the sequence (f_n) converges to the function 0 in the norm $\|\cdot\|_1$.

5. (a) Define what is meant by the statement that the normed space $(V, \|\cdot\|)$ is a Banach space.

Read your notes!

(b) Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two Banach spaces over the same field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Consider the mapping $N : V \times W \rightarrow \mathbb{R}$ defined by

$$N(v, w) := \|v\|_V + \|w\|_W \quad \forall (v, w) \in V \times W.$$

(i) Show that N is a norm on $V \times W$.

Trivial!

(ii) Show $V \times W$ is a Banach space when equipped with the norm N .

Let (v_n, w_n) be a Cauchy sequence in $(V \times W, N)$. We want to show that (v_n, w_n) converges in $(V \times W, N)$.

We have

$$\forall \varepsilon > 0 \quad \exists n_\varepsilon \in \mathbb{N} \text{ such that } n, m \geq n_\varepsilon \Rightarrow N((v_n, w_n) - (v_m, w_m)) < \varepsilon.$$

Since,

$$N((v_n, w_n) - (v_m, w_m)) = N((v_n - v_m, w_n - w_m)) = \|v_n - v_m\|_V + \|w_n - w_m\|_W,$$

we get that

$$n, m \geq n_\varepsilon \Rightarrow \begin{cases} \|v_n - v_m\|_V < \varepsilon, \\ \|w_n - w_m\|_W < \varepsilon. \end{cases}$$

Therefore, the sequences (v_n) and (w_n) are Cauchy in the Banach spaces $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ respectively. So, Both sequences converge in $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ some v and w respectively. Thus,

$$\begin{aligned}\lim_{n \rightarrow \infty} N((v_n, w_n) - (v, w)) &= \lim_{n \rightarrow \infty} N(v_n - v, w_n - w) = \lim_{n \rightarrow \infty} [\|v_n - v\|_V + \|w_n - w\|_W] \\ &= \lim_{n \rightarrow \infty} \|v_n - v\|_V + \lim_{n \rightarrow \infty} \|w_n - w\|_W = 0 + 0 = 0.\end{aligned}$$

6. Let

$$\ell^\infty(\mathbb{R}) := \{x = (x_n) \text{ sequence of real numbers such that } \sup_{n \in \mathbb{N}} |x_n| < +\infty\},$$

$$\ell^p(\mathbb{R}) := \left\{ x = (x_n) \text{ sequence of real numbers such that } \sum_{n=0}^{\infty} |x_n|^p < +\infty \right\} \text{ with } 1 \leq p < \infty.$$

(a) Say which of the following sequences belong(s) to $\ell^1(\mathbb{R})$:

$$(i) \quad x_n = \frac{2^n}{n + 9^{n/3}} \quad \forall n \in \mathbb{N};$$

We want to check whether $\sum_{n=0}^{\infty} |x_n| < \infty$.

Notice that

$$0 < x_n = \frac{2^n}{n + 9^{n/3}} \leq \frac{2^n}{9^{n/3}} = \left(\frac{2}{9^{1/3}}\right)^n.$$

Now, since $\frac{2}{9^{1/3}} < 1$ it follows that the geometric series

$$\sum_{n=0}^{\infty} \left(\frac{2}{9^{1/3}}\right)^n < \infty \quad \Rightarrow \quad \sum_{n=0}^{\infty} |x_n| < \infty \text{ (by comparison test).}$$

$$(ii) \quad \begin{cases} y_0 = 0 \\ y_n := \frac{1}{n} \arctan\left(\frac{1}{\sqrt{n}}\right) \quad \forall n \in \mathbb{N} \setminus \{0\}. \end{cases}$$

(Hint: $\arctan x \leq x \quad \forall x \geq 0$).

Also here, we want to check whether $\sum_{n=0}^{\infty} |y_n| < \infty$.

Using the hint $\arctan x \leq x$ we get that

$$0 \leq y_n \leq \frac{1}{n} \times \frac{1}{\sqrt{n}} = \frac{1}{n^{3/2}}.$$

Hence,

$$\sum_{n=0}^{\infty} \frac{1}{n^{3/2}} < \infty \quad \text{since } \frac{3}{2} > 1 \quad \Rightarrow \quad \sum_{n=0}^{\infty} |y_n| < \infty.$$

(b) Show that $\ell^p(\mathbb{R}) \subset \ell^q(\mathbb{R})$ for $1 \leq p \leq q \leq \infty$.

The case $q = \infty$. Let $x = (x_n) \in \ell^p(\mathbb{R})$.

$$\sum_{n=0}^{\infty} |x_n|^p < \infty \Rightarrow \lim_{n \rightarrow \infty} x_n = 0 \Rightarrow (x_n) \text{ is bounded} \Rightarrow x \in \ell^\infty(\mathbb{R}).$$

The case $q < \infty$. Let $x = (x_n) \in \ell^p(\mathbb{R})$. That is, $\sum_{n=0}^{\infty} |x_n|^p < \infty$. As before, we get $x \in \ell^\infty(\mathbb{R})$. That is, there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$. So, we get for $p < q$ that

$$\sum_{n=0}^{\infty} |x_n|^q = \sum_{n=0}^{\infty} |x_n|^p |x_n|^{q-p} \leq \sum_{n=0}^{\infty} |x_n|^p M^{q-p} = M^{q-p} \sum_{n=0}^{\infty} |x_n|^p < \infty \Rightarrow x \in \ell^q(\mathbb{R}).$$

7. We recall that

$$\|(x, y)\|_\infty := \max(|x|, |y|) \quad \text{and} \quad \|(x, y)\|_1 := |x| + |y| \quad \forall (x, y) \in \mathbb{R}^2.$$

For $n \in \mathbb{N}$, we set

$$u_n := \left(\frac{1}{1+n}, 1 + e^{-n} \right) \in \mathbb{R}^2.$$

(a) Study the convergence of the sequence (u_n) in $\mathbb{R}^2$ equipped with the norm $\|\cdot\|_\infty$.

(b) Study the convergence of the sequence (u_n) in $\mathbb{R}^2$ equipped with the norm $\|\cdot\|_1$.

Notice that, from the structures of the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$, the sequence (u_n) converges to $u = (x, y)$ in those norms if and only if $\lim_{n \rightarrow \infty} \frac{1}{n+1} = x$ and $\lim_{n \rightarrow \infty} (1 + e^{-n}) = y$. Hence, we find that $x = 0$ and $y = 1$. Therefore the sequence (u_n) converges to $u = (0, 1)$ in both norms.